Website URL: https://when-food-exists-but-hunger-persists.base44.app
GitHub URL: https://github.com/dnyanadab/DATA-MINING-PROJECT-GROUP-13

Milestone 3: Model Implementation

REGRESSION

Three regression models were selected to provide a comprehensive and robust analysis of the relationship between socioeconomic factors and low food access. Each model plays a distinct role:

Linear Regression (OLS)

Linear Regression serves as the baseline model. It is the simplest and most interpretable approach, making it ideal for establishing a performance benchmark. OLS directly quantifies how each predictor (e.g., income, poverty rate, infrastructure index) relates to the share of the population with low food access. However, because many socioeconomic predictors are likely correlated with each other, OLS may produce unstable coefficient estimates, a limitation addressed by the regularised models.

Ridge Regression

Ridge Regression introduces an L2 regularisation penalty that shrinks coefficient magnitudes without eliminating any predictors entirely. This is particularly appropriate when predictors are correlated (multicollinearity), a common condition in socioeconomic datasets where variables such as PovertyRate, MedianIncome, and UnemploymentRate overlap in meaning. Ridge improves model stability and generalisation while retaining all features for interpretation.

Lasso Regression

Lasso Regression applies an L1 penalty that can reduce some coefficients to exactly zero, effectively performing automatic feature selection. This is useful for identifying the most influential predictors and simplifying the model. Using all three models in combination allows for a balanced approach that compares interpretability, stability, and sparsity.

Model Assumptions

All three models are built upon the standard linear regression framework and therefore share the following assumptions:

- Linearity: The relationship between each predictor and the response variable is assumed to be linear. Non-linear relationships will not be fully captured.
- Independence of Errors: Observations (counties) are treated as independent. Spatial autocorrelation between neighbouring counties is a potential violation not corrected in this analysis.
- Homoscedasticity: The variance of residuals should be constant across all fitted values. Violations are assessed via the Residual Plot.
- Normality of Residuals: Residuals should follow a roughly normal distribution, which is assessed via the Residual Distribution histogram.
- No Perfect Multicollinearity: Predictors should not be perfectly correlated with each other.

Feature Selection & Hyperparameter Tuning

Ridge and Lasso require selecting a regularisation strength parameter alpha. GridSearchCV with 5-fold (Ridge) and 10-fold (Lasso) cross-validation was used to identify the optimal alpha from a predefined candidate grid:

- Ridge candidate alphas: [0.01, 0.1, 1, 10, 50, 100, 200] → Best alpha: 50
- Lasso candidate alphas: [0.0001, 0.001, 0.01, 0.1, 1, 10] → Best alpha: 0.0001

The high optimal alpha for Ridge (50) indicates that meaningful regularisation was needed, consistent with the multicollinearity flagged in the OLS output. The very low optimal alpha for Lasso (0.0001) indicates that Lasso applied minimal shrinkage and did not eliminate any features essentially behaving like standard OLS at this setting.

Features were standardised using StandardScaler before fitting Ridge and Lasso, as regularisation is sensitive to the scale of predictors.

Test Set Results

Model	R^2	MSE	RMSE
OLS (Linear)	0.1300	0.0304	0.1745
Ridge (α=50)	0.1342	0.0303	0.1741
Lasso (α=0.0001)	0.1309	0.0304	0.1744

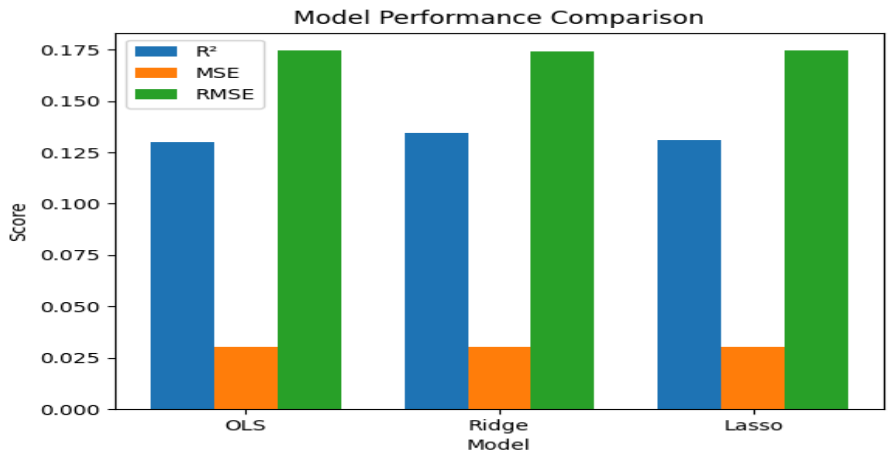
Cross-Validation Results (10-Fold, Training Set)

Model	CV Mean R^2	CV Std R^2	
OLS (Linear)	0.2179	±0.0840	

Ridge (α=50)	0.2186	±0.0821	
Lasso (α=0.0001)	0.2179	±0.0836	

The CV R^2 values (~0.22) are notably higher than the test R^2 (~0.13), indicating some overfitting to the training distribution. Ridge achieved the lowest variance across folds (±0.0821), confirming that regularisation improves stability.

Model Comparison Chart



Ridge Coefficient Analysis

The standardised Ridge coefficients reveal the relative importance and direction of each predictor:

Feature	Coefficient	Direction
FoodInfrastructureIndex	0.0450	Increases low access
RetailProxy	0.0403	Increases low access
MedianAge	0.0281	Increases low access
MedianIncome	-0.0258	Decreases low access
UnemploymentRate	-0.0221	Decreases low access
NoVehicleRate	-0.0204	Decreases low access
PovertyRate	-0.0193	Decreases low access
DG_per_10k	-0.0137	Decreases low access

Notably, the direction of some coefficients (e.g., PovertyRate negative) may seem counterintuitive. This reflects the multivariate nature of the model: when controlling for all other variables simultaneously, higher poverty in a county may correlate with more concentrated (urban) settlement reducing the share classified as low-access by distance.

Model Diagnostics

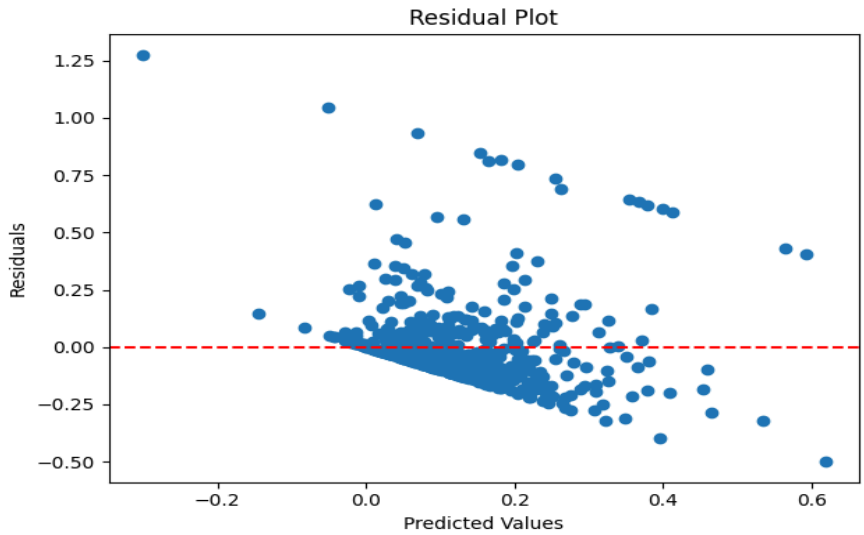


Figure 3: Residual Plot (Predicted vs. Residuals) — Ridge Model

The residual plot reveals a fan-shaped pattern, with residuals spreading more widely for lower predicted values. This indicates heteroscedasticity, the variance of errors is not constant. The concentration of predictions near zero suggests the model struggles with counties that have very high low-access shares, likely because those extreme cases involve non-linear dynamics not captured by a linear model.

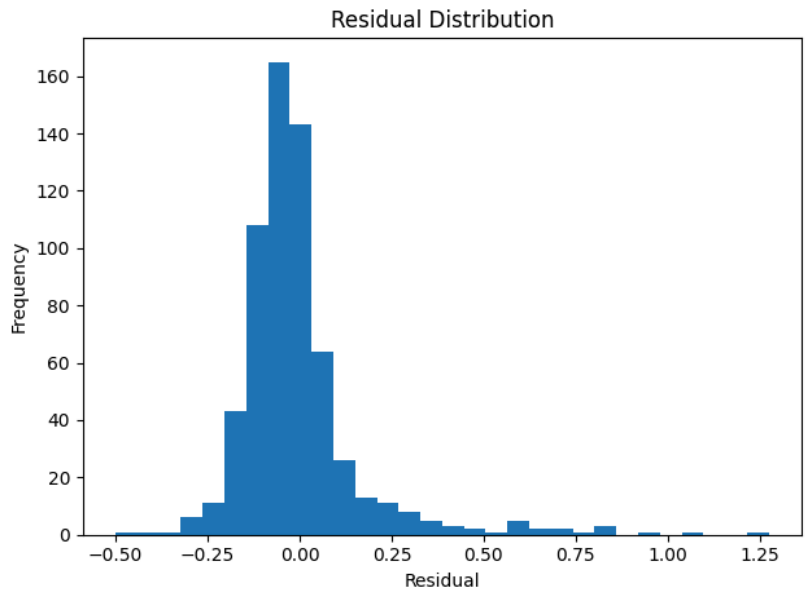


Figure 4: Residual Distribution — Ridge Model

The residual distribution is right-skewed with a heavy tail, deviating from the normality assumption. Most residuals cluster near zero, but a long positive tail reflects counties where the model significantly underestimates the low-access share.

Challenges Faced & Solutions

Challenge 1: Low Model Performance ($R^2 \approx 0.13$)

All three models explain only about 13% of test-set variance modest predictive power. This likely reflects non-linear relationships, spatial dependencies between neighbouring counties, and the bounded/zero-inflated nature of the target variable. The models selected important predictors but cannot capture more complex patterns.
Solution: The models are interpreted as establishing a valid linear baseline. Future work should explore non-linear models (e.g., Random Forests, Gradient Boosting) and spatial regression approaches that account for geographic clustering.

Challenge 2: Minimal Performance Difference Across Models

Ridge, Lasso, and OLS returned nearly identical RMSE and MSE values (all within 0.001 of each other), making model selection primarily a theoretical rather than empirical choice.
Solution: RMSE was used as the primary metric for consistency. Ridge was ultimately selected due to its slightly lower RMSE and higher cross-validation R^2 stability, as well as the theoretically sound rationale of addressing multicollinearity.

Challenge 3: Lasso Did Not Perform Feature Selection

The optimal Lasso alpha was extremely small (0.0001), meaning the L1 penalty was insufficient to shrink any coefficients to zero. Lasso effectively behaved like standard OLS for this dataset.
Solution: This result suggests that all 8 retained predictors contribute meaningful signal and that enforced sparsity is not appropriate here. Ridge, which retains all features while controlling their magnitude, is therefore the preferred regularised model.

Challenge 4: Assumption Violations

The residual diagnostics confirm violations of homoscedasticity and normality, two core assumptions of linear regression. The right-skewed residual distribution and heteroscedastic residual plot indicate the model is better suited to predicting typical counties than extreme outliers.
Solution: Assumption violations are documented and interpreted as an evidence-based motivation for future non-linear modelling. Residuals plots and distribution histograms were used for transparent assumption checking.

Challenge 5: Hyperparameter Sensitivity

Regularisation performance depends on selecting the correct alpha. A poor choice can lead to underfitting (alpha too high) or no effective regularisation (alpha too low).
Solution: GridSearchCV with k-fold cross-validation was applied for both Ridge and Lasso, ensuring robust, data-driven alpha selection rather than manual guessing.

Why Ridge Regression Performs Best

Ridge Regression was selected as the final model based on the following justification:

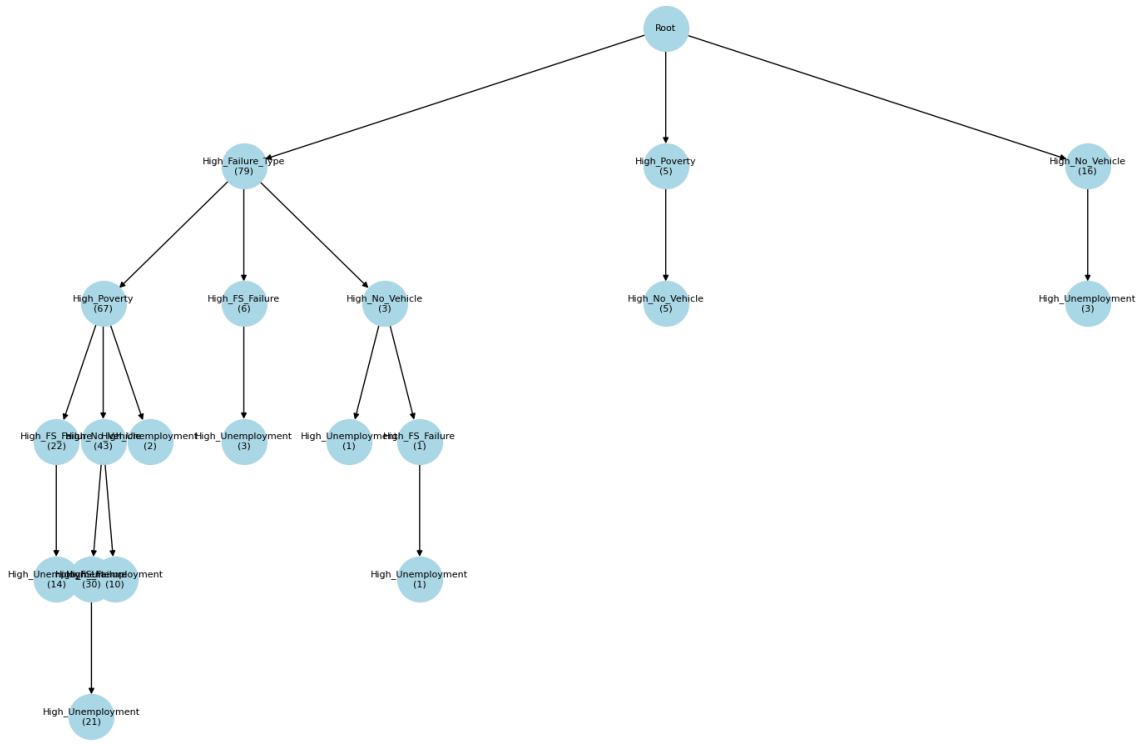
- Lowest test RMSE (0.1741) and MSE (0.0303): Ridge achieved the best predictive accuracy on unseen data, however marginal the difference.
- Lowest cross-validation variance (±0.0821): Ridge demonstrated the most stable generalisation across folds, indicating better robustness to variation in training data.
- Full feature retention: Unlike Lasso, Ridge retains all predictors, which is appropriate given that all 8 features were statistically significant in the OLS model.
- Lasso limitation: Lasso's near-zero alpha means it offered no practical advantage over OLS in this dataset, ruling it out as the preferred regularised approach.

Despite its marginal quantitative superiority, Ridge is preferred because it directly addresses the main structural weakness of OLS(multicollinearity), while maintaining full interpretability. The overall low R^2 across all models suggests that the most impactful future improvement would come from adopting non-linear approaches, not from choosing between these three linear variants.

FREQUENT PATTERN TREE AND APRIORI ALGORITHM

1) FREQUENT PATTERN TREE

Continuous variables (like PovertyRate and FoodSystemFailureScore) were binarized based on their medians into "High" and "Low" flags. Categorical variables were retained as distinct items. Minimum support of 0.2 was used to generate frequent itemsets.
Association rules were generated using lift as a measure having a minimum threshold as 1.0 FP Tree was created on a small sample of the dataset containing 50-100 values.



To visualize the **FP-Tree**, use a directed graph where each node represents an item and its frequency in the dataset.
Root Node : The starting point of all transactions.
Branches: Each path from the root represents a set of co-occurring features (e.g., a county having both High_Poverty and High_FS_Failure).
Node Labels: The text shows the item name and the count (frequency) of how many times that specific prefix path appears in the data.

Observe how several High Failure nodes are near the top. This shows that the High Failure trait is a common trait that other risk factors (like Unemployment or Lack of Vehicles) branch off from.

```
Frequent Itemset (Pattern) - Support
1) Type_High Failure - 91.2%
2) High_FoodSystemFailureScore, Type_High Failure - 50.0%
3) Low_NoVehicleRate - 51.1%
4) Low_UnemploymentRate - 50.8%
5) High_county_low_income_low_access, - 47.9%
High_county_low_access
```

1) High Failure & Access: If a county has a High Food System Failure Score AND High Low Access, there is a 96.3% confidence that it also has High Low Income / Low Access (Lift: 2.13).

2)Unemployment Link: Counties with High Unemployment and High Low Access are strongly associated with High Low Income / Low Access (Confidence: 97.9%).

3) Transportation: There is a strong pattern where Low No-Vehicle Rate (meaning higher car ownership) often appears in counties with High Food System Failure, suggesting that car ownership alone does not prevent a county from being classified as "High Failure."

2) APRIORI ALGORITHM

Key variables such as FoodSystemType, PovertyGroup, PovertyRate, UnemploymentRate, MedianIncome, FoodSystemFailureScore, and DG_per_10k (Dollar General stores per 10k people) were selected. Continuous variables were binned into three categories (Low, Medium, High) using quantiles to allow the algorithm to process them as categorical data. Frequent itemsets were identified with a minimum support of 0.1, and association rules were generated with a minimum confidence of 0.5

"High Failure" status has a base support of 91% across the dataset. This high prevalence means that almost any combination of negative indicators such as High Poverty or Low Income acts as a near-certain predictor of a failing food system. The rules highlight a specific shift in how Dollar General (DG) stores associate with food system health as their density increases:

- Medium DG Density: Shows a confidence of approximately 96.5% for system failure.

- High DG Density: While still highly associated with failure (93.2% confidence), these rules more frequently include UnemploymentRate_High.

This suggests that a Medium density of discount retail is already enough to signal a compromised food system. Further increases to High density are likely a secondary effect of labor market distress rather than a standalone driver of food access issues.

Observations

Socio-Economic Predictors of Poverty: There is a near-perfect association between a Very High Poverty Group classification and the combination of high poverty rates and low median income, validating the consistency of the group labels.

Food System Failure Risk: Counties with a High Food System Failure Score are almost certainly classified as having a High Failure Food System Type (Confidence: 1.0). This suggests the scoring model is highly consistent with the qualitative categorization.

The Dollar General Correlation: High densities of Dollar General stores (DG_per_10k_High) are strongly associated with High Failure food systems (93% confidence). This supports research suggesting these stores proliferate in areas with limited traditional grocery access.

Employment and Food Risk: High unemployment combined with high Dollar General density further strengthens the link to a High Failure food system classification (Lift: 1.03).

Classification Analysis

The intent of the classification analysis was to incorporate models that are most associated with classification to understand the drivers behind food access. Popular classifiers, typically decision tree models, capture non-linear relationships effectively which brings valuable insight to our analysis that other methods cannot easily replicate.

To facilitate this, we categorized the “FoodSystemFailure” variable by labeling counties as “HighFailure” who were at or above the 80th percentile of scores. Counties that were below this threshold were labeled as “Normal”. Thus our target class was morphed into a binary label of “High” or “Normal” failure. This reduces the complexity of the final model while also maintaining a directional relationship. Lastly, the 80th percentile decision boundary was chosen based off of an unsupervised 1d clustering analysis in which the 80th percentile captured the most variance between the two groups.

Model Selection

We chose three decision tree classifiers based on the assumption that they would capture non-linear relationships in a way that our other methods may not be able to capture.

Baseline Model : Random Forest

- Random Forest is a tree-based algorithm that simultaneously creates multiple deep decision trees for either classification or regression tasks.
- Random Forest was chosen as the baseline tree-based model due to the ease of training, resistance to overfitting, and straightforward hyperparameter tuning.

Experimental Model 1 : CatBoost

- CatBoost is a Gradient Boosted Decision tree model that creates multiple shallow trees which are iteratively adjusted using a gradient boosting technique.
- We chose CatBoost as an experimental model due to its ability to create more accurate models out of the box with the tradeoff of higher training complexity and overfitting issues.

Experimental Model 2: XGBoost

- XGBoost is a Gradient Boosted Decision Tree model that creates multiple shallow trees which are iteratively adjusted using a gradient boosting technique. The difference between this model and CatBoost is that XGBoost needs more time to be trained as there are far more hyperparameters to be adjusted.
- We chose XGBoost as the second experimental model to see if there were any significant differences between the two most popular gradient descent models available and if they would generate separate insights.

Training Process

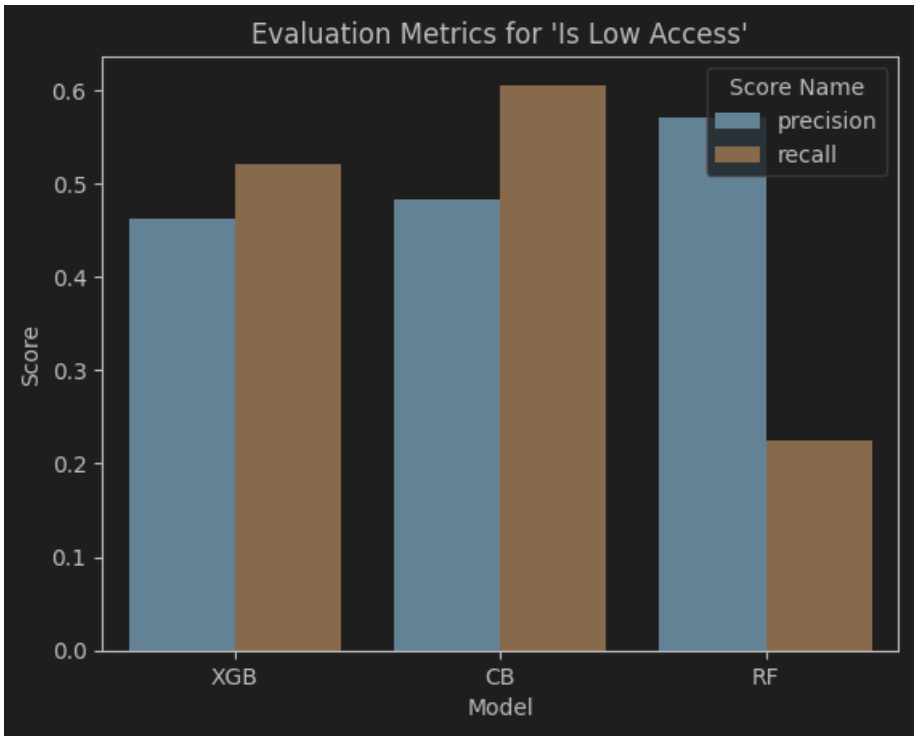
We utilized KFold (k=5) cross validation for testing the model during hyperparameter tuning as well as using a 10% holdout dataset for best model evaluation. Due to computational limitations, we used a shallow set of hyperparameters to tune during grid search which were suggested by AI. The hyperparameters that were changed and the training time of each model was the following:

- **Random Forest**
    - Training Time: 40s
    - HP Tuned: min_features, min_samples_leaf, max_depth
- **Cat Boost**
    - Training Time: 5min
    - HP Tuned: depth, learning_rate, iterations, l2_leaf_reg
- **XGBoost**
    - Training Time: 2min

- HP Tuned: max_depth, learning_rate, n_estimators, colsamples_bytree

Evaluation

Although the accuracy of the model is not the primary objective of the analysis, a model that is the most predictive of the output should yield the most accurate insights about the underlying dataset. To accommodate this, we chose recall and precision as our performance metrics so that the chosen model did the best job differentiating between the high and low access groups regardless of sample size. To assess precision and recall, we used a 10% holdout set that was not used in the model training.

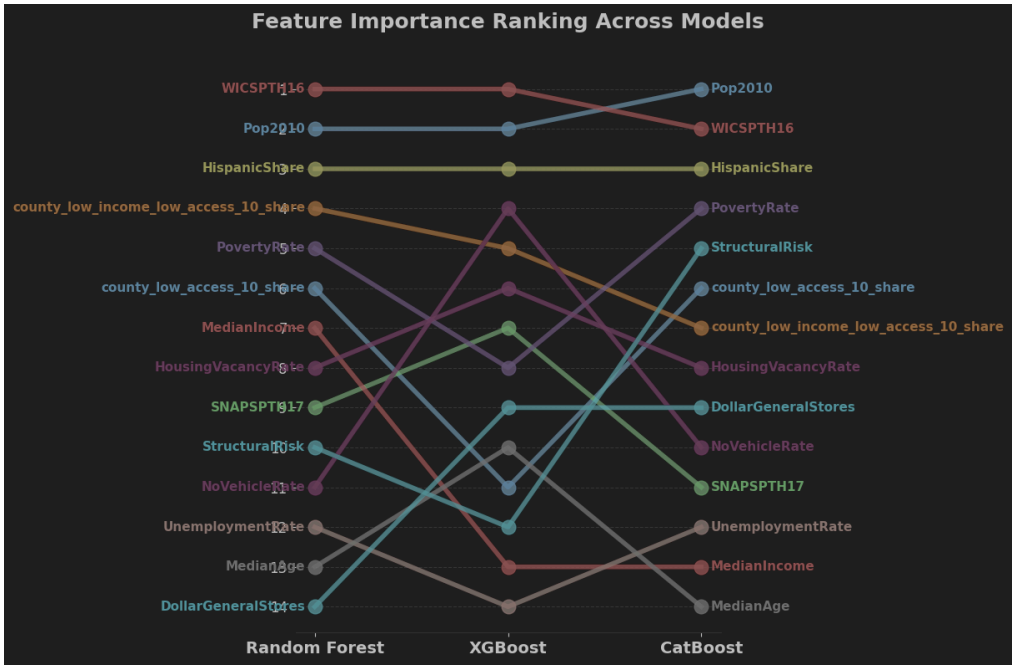


From the above graph, the CatBoost model had the highest recall and the second highest precision score. This is expected as CatBoost does a good job classifying data with minimal parameter tuning while XGBoost would have benefited from a more thorough grid search.

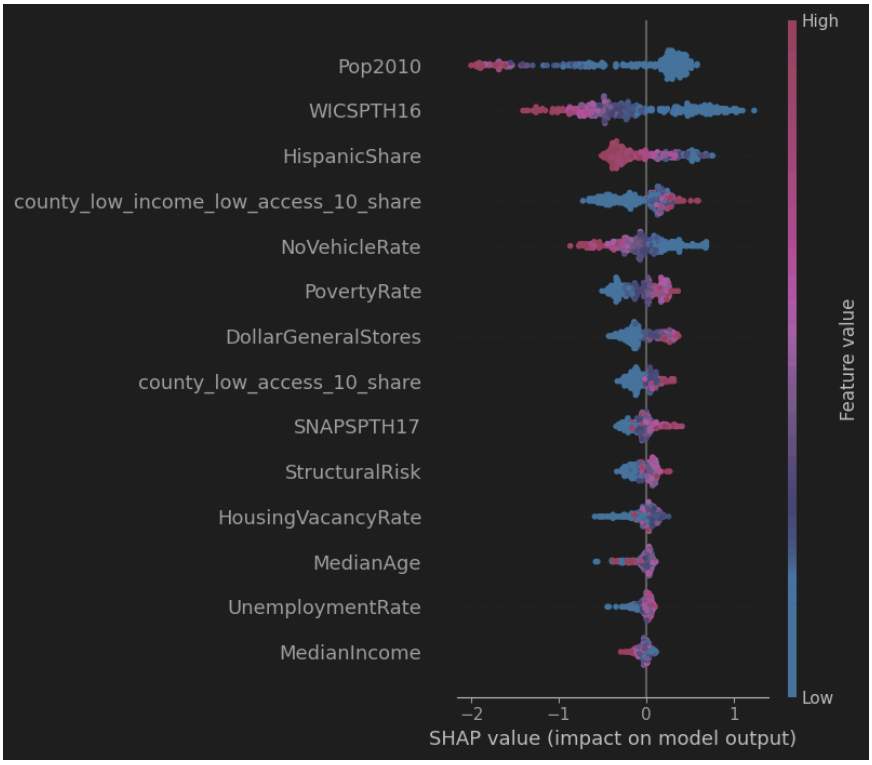
Insights

The following graph shows how each model interpreted the data using the “Feature Importance” metric at the end of training. As expected, the most predictive features were population(Pop2010) and grocery store access (WICSPTH16). What was interesting was that “hispanic share” was in the top three most predictive variables. Although directionality is not mentioned here, the predictability of HispanicShare could be explained by its relationship to other variables not covered in this dataset.

The scores that changed the most from model to model are NoVehicleRate, DollarGeneralStores, and MedianIncome.



For the most predictive model, CatBoost, we utilized a method called “SHAP” analysis that explains variability in any tree based model by measuring individual predictors against one another and measuring the output. The following graph reveals the directionality of the input variables and their relative impact on the model output.



As shown in the importance graph, Population seems to be a top predictor. However this analysis reveals that counties with extremely low populations have the least access. This tracks with our expectation that rural communities are at higher risk to low food access due to a lack of infrastructure. Moreover, the Dollar General Stores relationship shows that more stores in a county is associated with lower food access. Though not conclusive evidence, dollar general may be profiting off of food deserts.

Clustering Analysis: Structural Characterization of Food Access Inequality

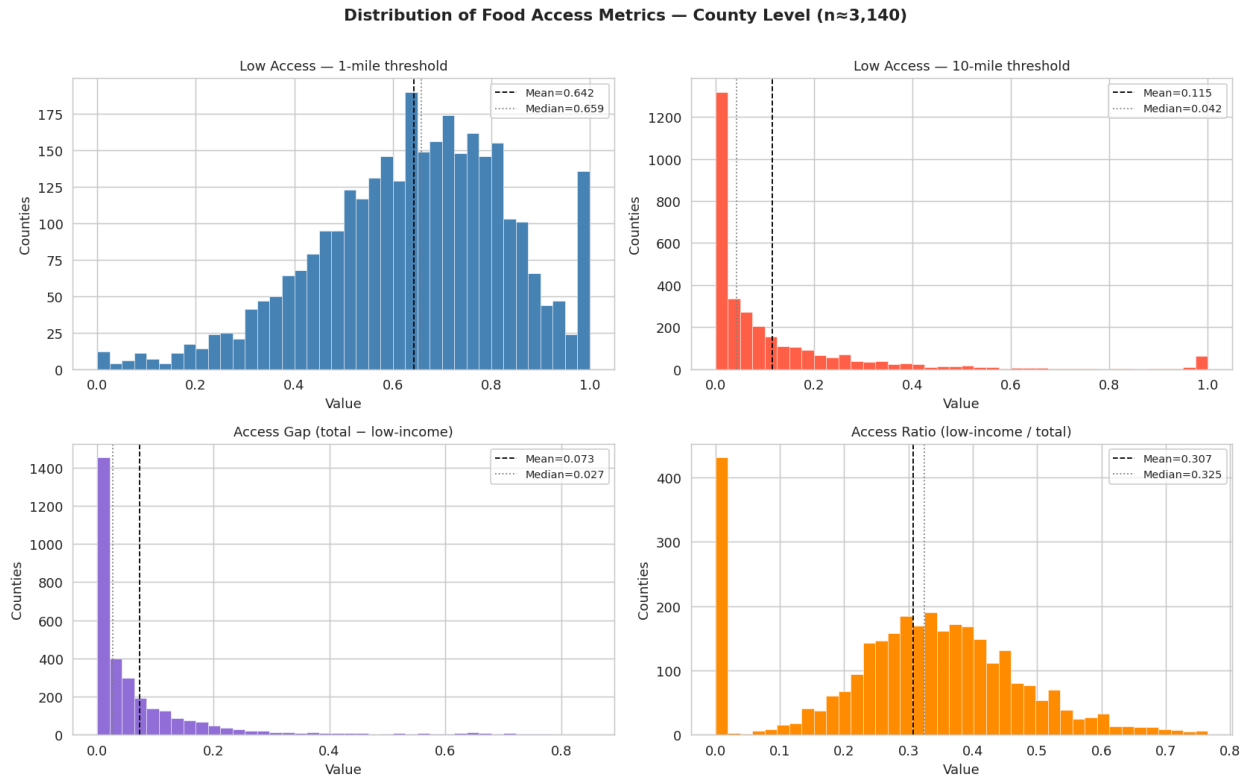
In this report, we describe how an analysis of Unsupervised Learning will help to understand the Hidden Structural Patterns of County-Level Food Access in the US. The analysis draws from a compilation of Food Access, American Community Survey (ACS), Food Environment, Dollar General, and is part of a larger analytical pipeline including a structural analysis through Supervised Learning Models defined by the uses of different types of regression and classification modeling.

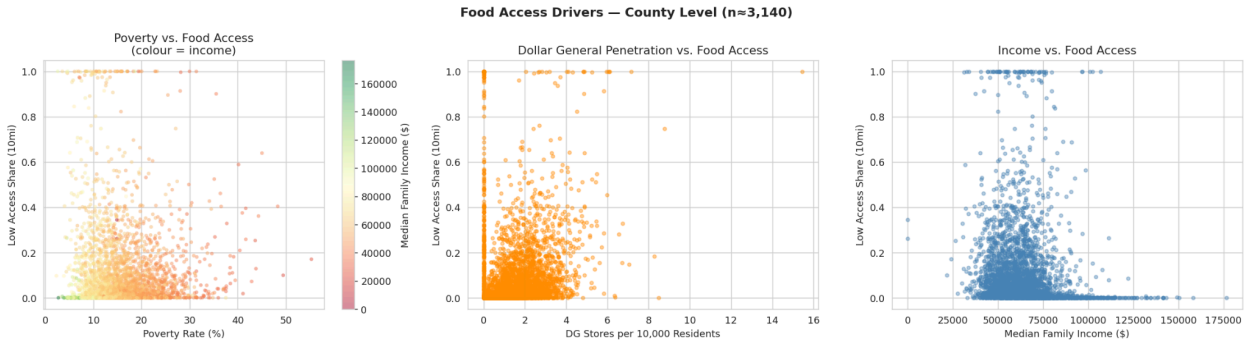
1. Analytical Objective

The purpose of clustering is to determine whether there is a natural grouping of counties based on multi-dimensional factors of Food Access. Because clustering methods do not use set outcomes like regression and classification modeling do, they uncover the inherent structure of the data.

2. Feature Space and Preprocessing

For all features used for clustering, the states were created using analysed variables that represent physical access (i.e., low_access_1, low_access_10), economic (PovertyRate and MedianFamilyIncome), coupled vulnerability (low_income_access_10), as well as other engineered variables (like access_gap, access_ratio, DG_per_10k, food_desert_risk), and others. Additionally, robust scaling was used to eliminate the impact of outliers. The data set was reduced in dimension using PCA, while UMAP was only used to produce a visual of the results of clustering.



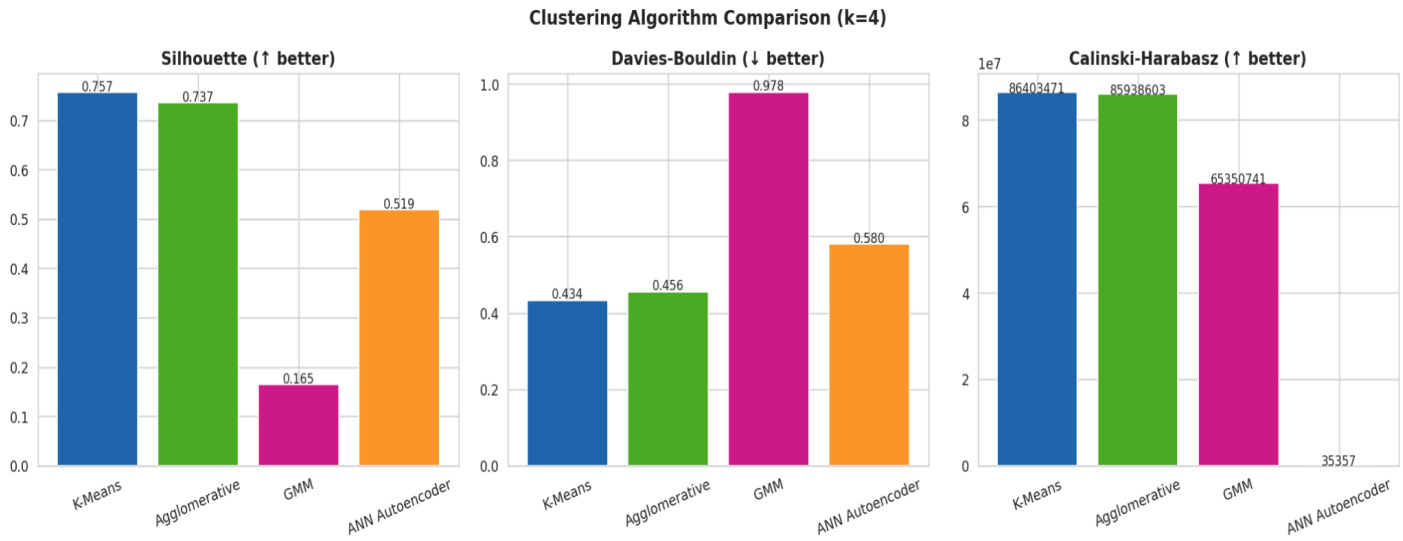


3. Clustering Models

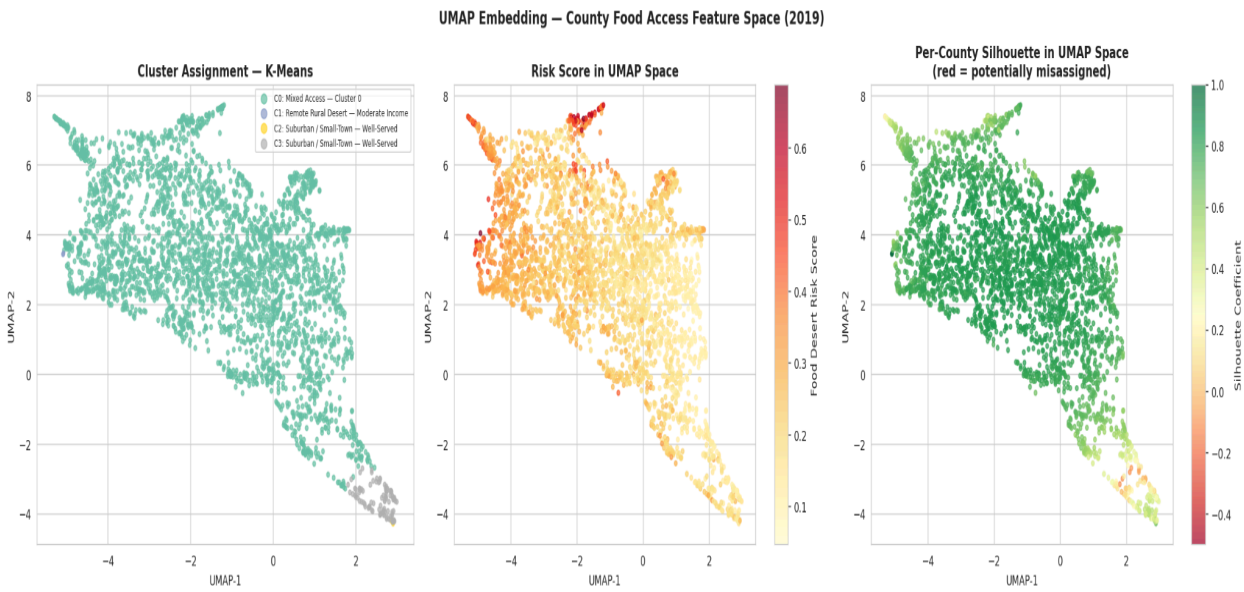
Several different clustering algorithms were explored when clustering counties through K-Means, Agglomerative Clustering, Gaussian Mixture Models, and Autoencoder-Based Clustering. The robustness of each of the models used for clustering was reviewed based on the assumptions made about the shape, distribution, and linearity of the clusters formed.

4. Evaluation of Model

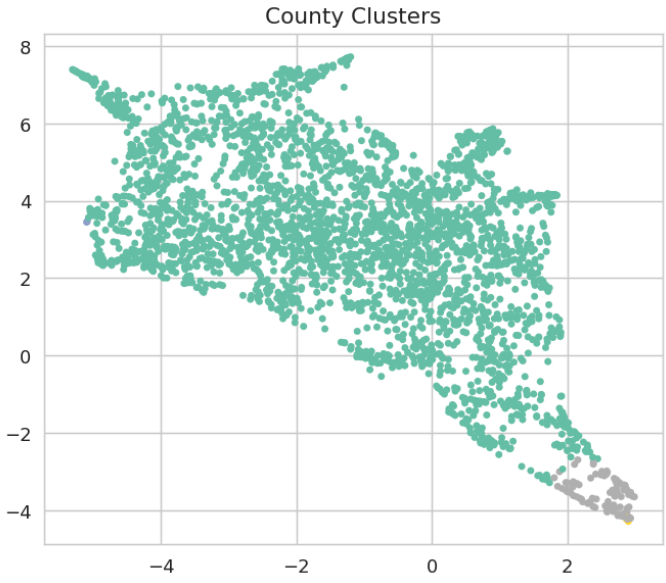
The silhouette score produced by K-Means Clustering was highest at approximately 0.75. There was a good separation within the feature space generated by K-Means. The results generated by Agglomerative Clustering also yielded relatively similar results. On the other hand, GMM and Autoencoders were provided with inferior results due to incorrect assumptions as well as overfitting the dataset.



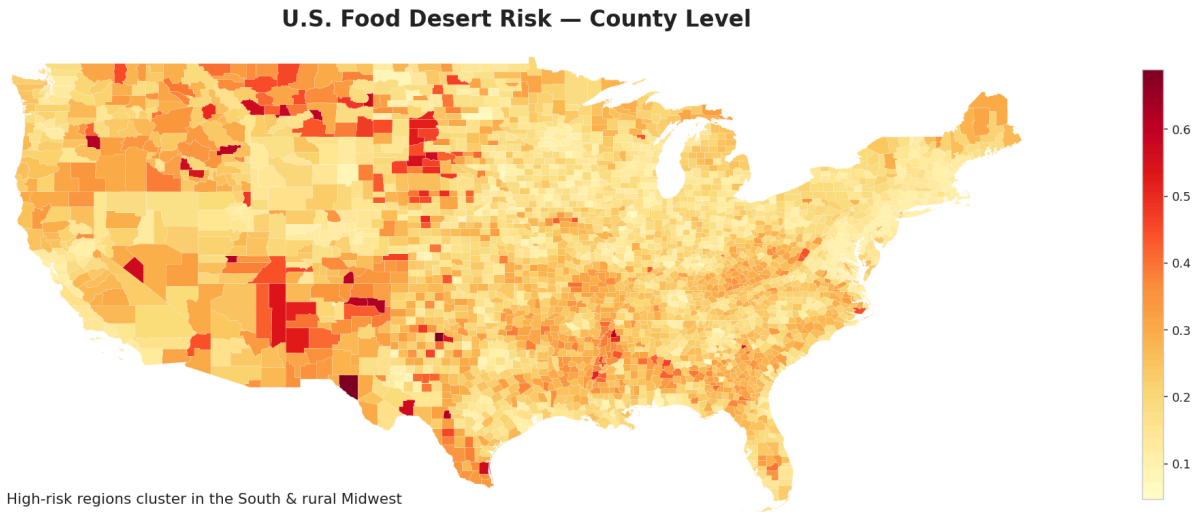
UMAP visualization indicates that counties in the central cluster share many repeating structural characteristics and that only a small number of counties belong to their own separate clusters. The majority of counties appear to be represented by a more continuous distribution, and very few are clearly separated from one another (by geographical distance).

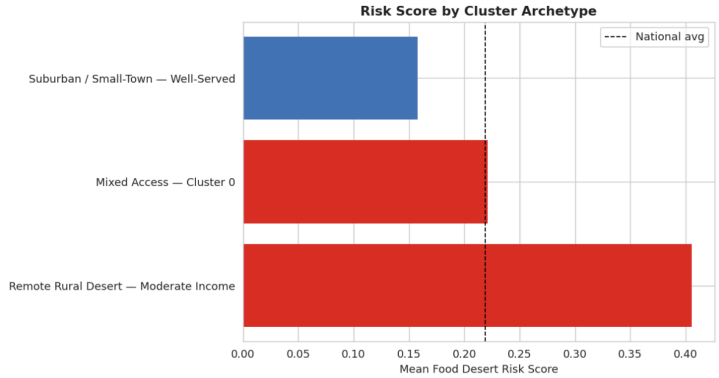


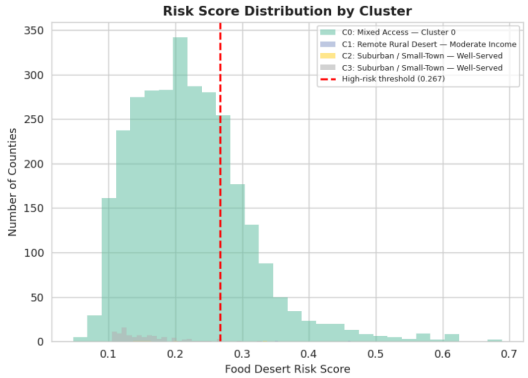
Cluster maps demonstrate that the outlier clusters tend to be concentrated in the rural areas of the Midwest, southwest, and southeastern USA, and that there is a clear geographic distribution for the clustering of food deserts. Therefore, even with one large cluster, the clusters are producing regions with meaningful regional differences.



Risk maps for food desolation indicate that risk levels clearly follow a geographic gradient from high risk (southern counties) to low-risk (more northern counties) areas. In addition to verifying the composite risk measure produced for food desolation, the geographic distribution of risk provides additional data supporting the previously known socioeconomic differentialNots.



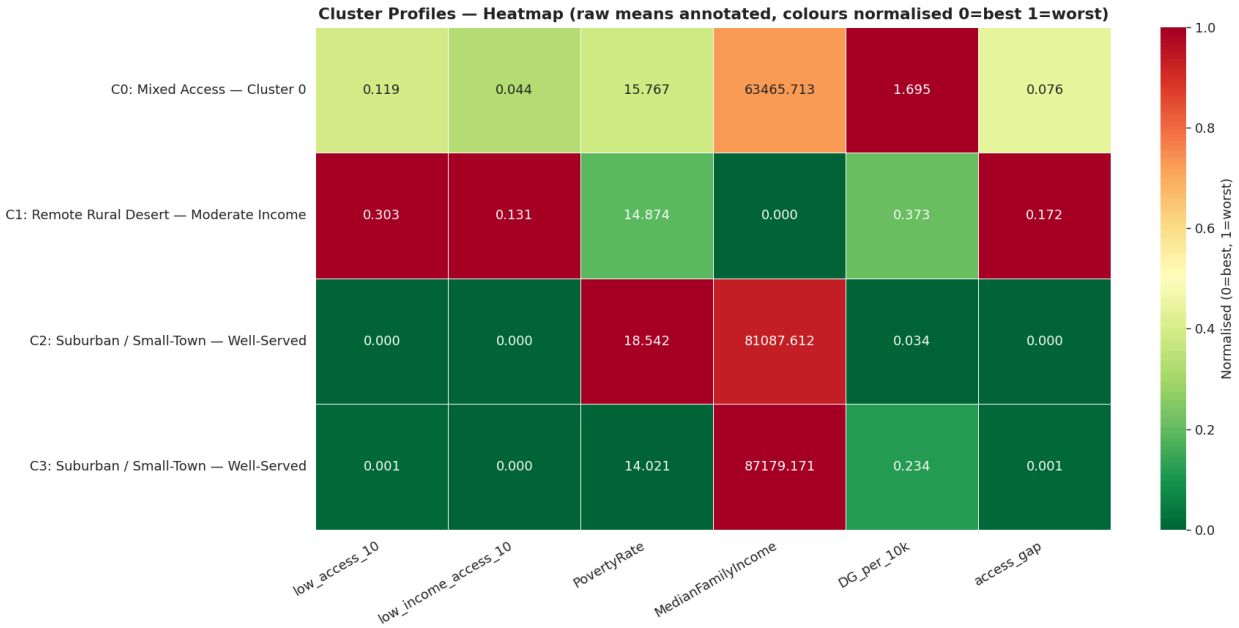


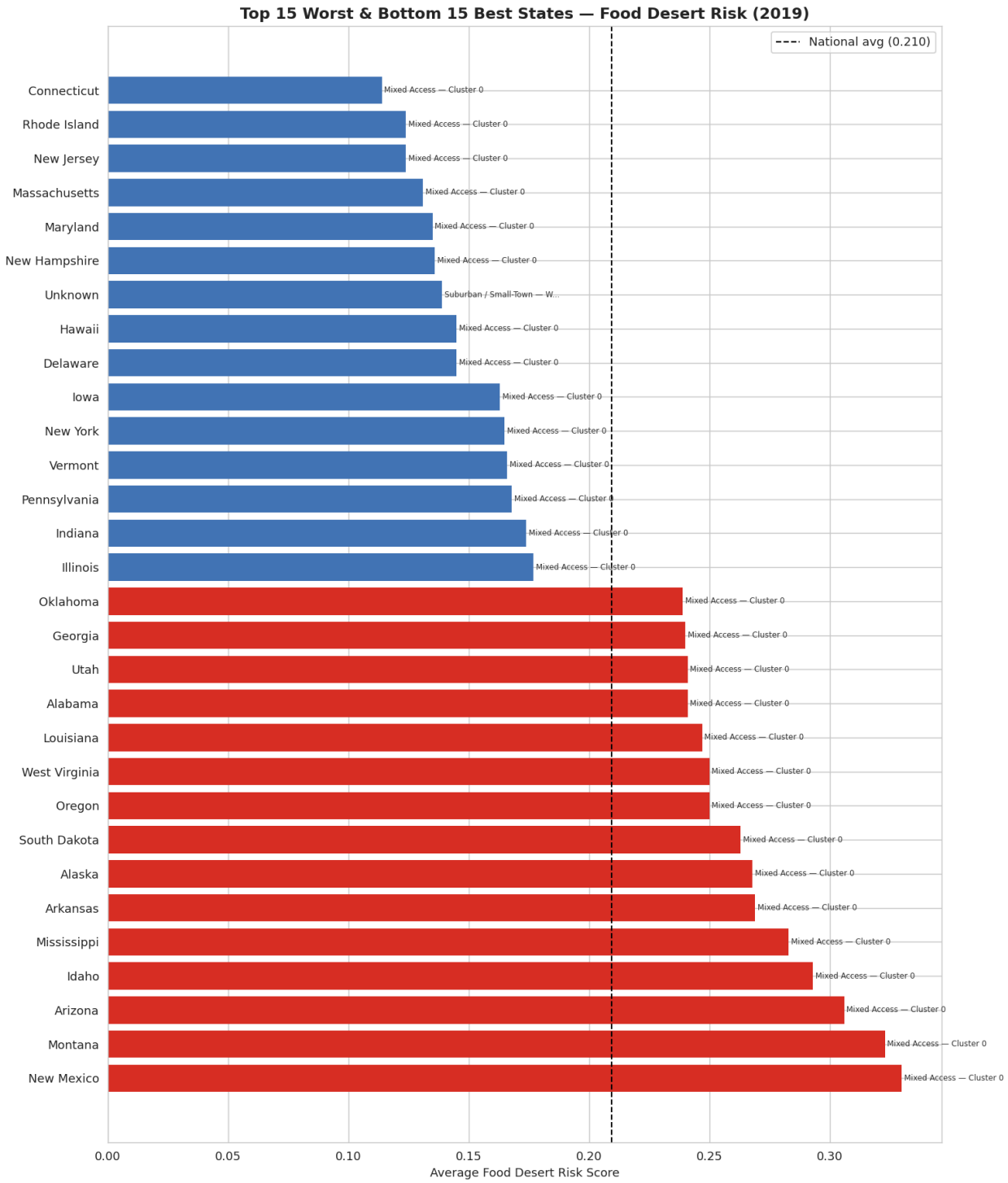


Socioeconomic variables are primarily responsible for explaining food access and include income and poverty measures. Thus, food access inequality is a structural issue, and socioeconomic variables are the leading explanatory variables in addressing that issue.

5. Findings from Clustering

The results of clustering show that food access at the county level is a continuum that is defined by one continuous socio-economic gradient. There are some identifiable extreme clusters, but most counties will not cluster into separate groups or categories. Therefore, policymakers need to account for gradual changes within a continuum rather than distinct categories when implementing policy solutions to address food access.


Cluster Profiles — Heatmap (raw means annotated, colours normalised 0=best 1=worst)
C0: Mixed Access — Cluster 0
C1: Remote Rural Desert — Moderate Income
C2: Suburban / Small-Town — Well-Served
C3: Suburban / Small-Town — Well-Served
0.119
0.044
15.767
63465.713
1.695
0.076
0.303
0.131
14.874
0.000
0.373
0.172
0.000
0.000
18.542
81087.612
0.034
0.000
0.001
0.000
14.021
87179.171
0.234
0.001
low_access_10
low_income_access_10
PovertyRate
MedianFamilyIncome
DG_per_10k
access_gap
Normalised (0=best, 1=worst)
1.0
0.8
0.6
0.4
0.2
0.0



6. Overall Implications of the Findings from Clustering

The analysis of clustering presents a structural view of the degree of inequality in food access based on geographic and socio-economic factors and the amount of commonality among food distributions. Clustering results support the predictive models discussed in previous sections of this report, and recommend that features that lack defined groups with a good deal of modification should be used in unsupervised learning.

Within Each Method:

- **Regression → Ridge Regression** was selected as best, with the lowest test RMSE (0.1741) and the most stable cross-validation variance (±0.0821). That said, all three regression models performed nearly identically, and the overall R^2 of ~0.13 means they only explained about 13% of variance — a weak result.
- **Classification → CatBoost** was the strongest, achieving the highest recall and second-highest precision among Random Forest, XGBoost, and CatBoost, with minimal hyperparameter tuning needed.
- **Clustering → K-Means** came out on top with the highest silhouette score (~0.75) and the best separation in feature space.

Overall Best Model for the Research Goals:

CatBoost is arguably the most valuable model for this research.

1. The core research goal is understanding *drivers* of food access inequality, not just predicting a number. CatBoost's SHAP analysis provided the clearest directional insights about which variables matter and *how*.
2. It captured non-linear relationships that the regression models fundamentally couldn't, which the document explicitly flags as a major limitation of Ridge/OLS.
3. It identified practically meaningful findings — like rural population and grocery store access being top predictors, and the Dollar General density correlation with food deserts.

The regression models serve as a useful **linear baseline**, and K-Means clustering provides valuable **structural context**, but CatBoost delivers the most actionable and interpretable insight for the research question.